Overlap and Balance in Causal Inference

Yue Wu

March 14, 2026

Overlap assumption

- weak overlap

$$0 < P(Z = 1 \mid X) < 1$$

- strict overlap
  For some constant $\eta \in (0, 0.5)$,

$$\eta \leq e(X) \leq 1 - \eta$$

- Theoretically, Overlap assumption is a must for the identification of ATE and Strict Overlap is needed for statistical analysis in many papers.

Overlap in IPW

(Wager 2024)

- oracle IPW estimator $\hat{\tau}^*_{ipw} = \frac{1}{n}\sum_{i=1}^{n}(\frac{Z_i Y_i}{e(X_i)} - \frac{(1-Z_i)Y_i}{1-e(X_i)})$ is asymptotically normal

$$\sqrt{n}(\hat{\tau}^*_{ipw} - \tau) \sim N(0, V^*_{IPW})$$

where the asymptotical variance

$$V_{IPW^*} = \mathrm{Var}\left[\tau(X_i)\right] + \mathbb{E}\left[\frac{\left(\mu_{(0)}(X_i) + (1-e(X_i))\tau(X_i)\right)^2}{e(X_i)(1-e(X_i))}\right] + \mathbb{E}\left[\frac{\sigma^2_{(1)}(X_i)}{e(X_i)} + \frac{\sigma^2_{(0)}(X_i)}{1-e(X_i)}\right]. \quad (1)$$

- all moments need to be well-defined and finite
  - if potential outcomes are uniformly bounded, it means we need $E(\frac{1}{e(X)(1-e(X)})) < \infty$
  - if the potential outcomes have finite second moments, it means we need bounded propensity, i.e. strict overlap

Overlap in PLM

Consider Partial linear model

$$E(Y \mid X, Z) = \beta^T Z + \theta(X)$$

i.e.

$$Y - E(Y|X) = \beta(Z - E(Z|X)) + U, \; E(U|X, Z) = 0$$

In causal settings, given unconfoundedness, $E(Y(1)|X) = \beta + \theta(X)$, $E(Y(0)|X) = \theta(X)$, ATE = β(our estimand).

- (Robinson 1988) (generalized FWL theorem) regression-of-residuals estimator is consistent $\sqrt{N}(\hat{\beta} - \beta) \xrightarrow{d} N(0, \sigma^2 \Phi^{-1})$. if
  - $\Phi = E[(Z - E(Z|X))(Z - E(Z|X))^\top]$ is positive definite.
    - $\Phi^{-1} = \frac{1}{E(e(x)(1-e(x))} = \frac{1}{E(Var(Z|X))}$ (convergence rate & overlap)
  - kernel estimators for first step, other ..

Overlap in R-learner

Considering HTE, now the PLM model becomes:

$$E(Y \mid X, Z) = \beta(X)Z + \theta(X)$$

i.e.

$$Y - m(X) = \beta(X)(Z - e(X)) + U$$

In causal settings, given unconfoundedness, $E(Y(1)|X) = \beta(X) + \theta(X)$, $E(Y(0)|X) = \theta(X)$, CATE $= \beta(X)$ (our estimand).

- (Nie and Wager 2021) equivalently

$$\tau^*(\cdot) = \arg\min_{\tau} \mathbb{E}\left[\left(Y - m^*(X) - (W - e^*(X))\tau(X)\right)^2\right]$$

  therefore estimator(minimize R-loss, cross fitting):

$$\hat{\tau} = \arg\min_{\tau} \left(\hat{L}_n(\tau) + \Lambda_n(\tau)\right)$$

  - penalized kernel R-learner is near oracle(depends on the local robustness property of R-loss) if
    - strict overlap: turning regret bound into MSE bound

$$(1-\eta)^{-2}R(\tau) < E[\{\tau(X_i) - \tau^*(X_i)\}^2] < \eta^{-2}R(\tau)$$

    - RKHS, good estimate of nuisance functions, kernel, ...
  - when $e(X) \approx 0$, R-learner roughly equivalent to X-learner(?)

Overlap in CRF - high-dimensional/heterogeneity

An example of strict overlap in high-dimension case.

- Bias of estimation(Wager and Athey 2018)

$$|\mathbb{E}[\hat{\tau}(x)] - \tau(x)| \lesssim 2Md\left(\frac{\eta s}{2k-1}\right)^{-\frac{1}{2}\frac{\log((1-\alpha)^{-1})}{\log(\alpha^{-1})}\frac{\pi}{d}} = \mathcal{O}\left(s^{-\frac{1}{2}\frac{\log((1-\alpha)^{-1})}{\log(\alpha^{-1})}\frac{\pi}{d}}\right)$$

  - bounded covariates $X_1, \ldots X_s \in [0,1]^d$
  - M-bounded and Lipschitz continuous conditional mean functions
  - strict overlap: for some $\eta > 0$ and all $x \in [0,1]^d$, $\eta < e(x) < 1 - \eta$.
  - A tree predictor grown by recursive partitioning is α-regular
- Tree beats NN under settings of low-dimensional signal structure[1]
- More
  - is strict overlap plausible for tree? how does the balance of data influence the structure(then the performance) of the tree?
  - could match/balance based weighting be used in CRF? (e.g. a balance-aware way of sample weighting/splitting[2])

[1] a data-driven way to determine which nearby observations receive more weight

[2] "An important challenge for future work is to design splitting rules that can automatically choose which characteristic of the training data to split on."

'overlapness' in population and sample sense

- why poor overlap?
  -
- poor overlap, then?
- when does that matter

overlap & balance

Define P_1 by $\frac{dP_1}{dP} = \frac{e(X)}{E_p(e(X))}$ and P_0 by $\frac{dP_0}{dP} = \frac{1-e(X)}{E_p(1-e(X))}$, which are the distribution of covariates in treatment and control arm. Let $P(Z=1) = \pi$ be a constant $\pi \in (0,1)$.
By Bayes' theorem, we have density ratio(a measure of balance) with propensity ratio(a measure of overlap) as

$$\frac{dP_1}{dP_0} = \frac{P(X|Z=1)}{P(X|Z=0)} = \frac{P(Z=1|X)}{P(Z=0|X)}\frac{P(Z=0)}{P(Z=1)} = \frac{P(Z=1|X)}{P(Z=0|X)}\frac{1-\pi}{\pi}$$

and the mean discrepancy(a measure of balance) is

$$E(X|Z=1) - E(X|Z=0) = E_{P_1}(X) - E_{P_0}(X)$$
$$= E_P(X(\frac{dP_1}{dP} - \frac{dP_0}{dP})) = E_P(X(\frac{e(X)-\pi}{\pi(1-\pi)}))$$

Also can derive other f-divergence form of distance to measure the difference of the two distributions.

$$D_f(Q_1 \| Q_0) := E_{Q_0}[f(\frac{dQ_1}{dQ_0})]$$

strict overlap & balance in high-dimensional setting

high-dimensional: (1) many covariates; (2) complex interactions among covariates in outcome model or propensity model

- Strict overlap with η implies bounded density ratio.

$$b_{min} =: \frac{1-\pi}{\pi}\frac{\eta}{1-\eta} \leq \frac{dP_1}{dP_0} \leq \frac{1-\pi}{\pi}\frac{1-\eta}{\eta} := b_{max}$$

- Strict overlap restricts global discrepancies between P_0 and P_1([support same?]), and these restrictions become more restrictive as the dimension of covariates grows large(D'Amour et al. 2021)[3]

$$D_f(P_1 \| P_0) \leq B_f(b_{min}, b_{max})$$

$$D_f(P_0 \| P_1) \leq B_f(b_{max}^{-1}, b_{min}^{-1})$$

- From data-generating view, if we have high-interaction confounding, then poor high-dimension overlap [?]

[3] where the upper bound B_f is actually free of p, while the left side may not. So strict overlap is very 'strict' in high-dimensional setting.

An example in data-generating view

Look into the classical setting:

$$e(X) = exp(W_e),\ X \sim N_p(\mu, \Sigma),\ W_e = \beta_0 + \sum_{j=1}^{p} \beta_j X_j \sim N(\beta_0 + \beta^T \mu, \beta^T \Sigma \beta)$$

Does strict overlap hold? for a constant $\eta \in (0, 1)$.
What would influence the divergence $D_f(P_1 || P_0)$?

measure of population overlapness

Overlapness can be a overall concept, or a local concept.

- $min(\eta)$ of strict overlap assumption $\eta < e(X) < 1 - \eta$
- $E(e(x)(1 - e(x))$?
- (Liu and Li 2025) Let $f_z(u) = Pr(e(X) = u \mid Z = z)$, the Bhattacharyya coefficient

$$\phi \equiv \int_0^1 \sqrt{f_1(u) f_0(u)}\, \mathrm{d}u.$$

  - assume propensity function is canonical a logit-normal distribution(by Lyapunov CLT), which can be approximated by Beta distribution
  - (r, ϕ) is sufficient to define this beta dist
  - given r, ϕ is monotonically increasing in both a and b.
- (Kim et al. 2025) Let $\Sigma_z = \mathbb{E}[\phi(x) \otimes \phi(x)\mathbf{1}(Z = z)]$, define weak overlap[?]

$$\Sigma_0 \preceq R\left(\Sigma_1 + \frac{\xi}{n}\mathbf{I}\right) \quad \text{and} \quad \Sigma_1 \preceq R\left(\Sigma_0 + \frac{\xi}{n}\mathbf{I}\right)$$

measure of sample balance

Balance is an overall concept(no local version), and we need a metric to describe the distance of two distributions.

- one-dimensional norm(e,g. overlap region of propensity score[4]); mean-matching/pair-matching (Rubin 1973)
- Mahalanobis distance[5] (which also contain terms that are functions of the covariates like power/interactions)
- p-values form KS test and paired t-tests (general because the same scale) (Diamond and Sekhon 2013)
- combine ps-score and original X and ..(better firstly orthogonalize by regressing)

[4] for ps score, better to use the linear predictor because it is note close to 0 or 1 (Diamond and Sekhon 2013)

[5] MD does not perform well when covariates have nonellipsoidal distributions?

Overlap-related methods

- Overlap condition influences the performance of estimators (Busso, DiNardo, and McCrary 2014)

? (Crump et al. 2009) [ATE] Characterize(by ps score) optimal subsamples for which the average treatment effect can be estimated most precisely.

we also consider the problem without imposing this restriction. Formally, we choose an estimand $\tau_{S,\mathbb{A}}$ by choosing the set $\mathbb{A} \subset \mathbb{X}$ that minimizes

$$V_{\mathbb{A}}^{\text{eff}} = \frac{1}{\{E(1_{X\in\mathbb{A}})\}^2} E\left[1_{X\in\mathbb{A}}\left\{\frac{\sigma_1^2(X)}{e(X)} + \frac{\sigma_0^2(X)}{1-e(X)}\right\}\right].$$

Defining $q(\mathbb{A}) = \text{pr}(X \in \mathbb{A}) = E(1_{X\in\mathbb{A}})$, we can write the objective function as

$$V_{\mathbb{A}}^{\text{eff}} = \frac{1}{q(\mathbb{A})} E\left\{\frac{\sigma_1^2(X)}{e(X)} + \frac{\sigma_0^2(X)}{1-e(X)} \middle| X \in \mathbb{A}\right\}.$$

We seek $\mathbb{A} = \mathbb{A}^*$, which minimizes $V_{\mathbb{A}}^{\text{eff}}$ among all closed subsets $\mathbb{A} \subset \mathbb{X}$.

- (Li, Thomas, and Li 2019) [ATE] Overlap Weighting

references

Busso, Matias, John DiNardo, and Justin McCrary (2014). "New evidence on the finite sample properties of propensity score reweighting and matching estimators". In: *Review of Economics and Statistics* 96.5, pp. 885–897.

Crump, Richard K et al. (2009). "Dealing with limited overlap in estimation of average treatment effects". In: *Biometrika* 96.1, pp. 187–199.

D'Amour, Alexander et al. (2021). "Overlap in observational studies with high-dimensional covariates". In: *Journal of Econometrics* 221.2, pp. 644–654.

Diamond, Alexis and Jasjeet S Sekhon (2013). "Genetic matching for estimating causal effects: A general multivariate matching method for achieving balance in observational studies". In: *Review of Economics and Statistics* 95.3, pp. 932–945.

Kim, Seok-Jin et al. (2025). "Transfer learning of cate with kernel ridge regression". In: *arXiv preprint arXiv:2502.11331*.

Li, Fan, Laine E Thomas, and Fan Li (2019). "Addressing extreme propensity scores via the overlap weights". In: *American journal of epidemiology* 188.1, pp. 250–257.

Liu, Bo and Fan Li (2025). "Sample size and power calculations for causal inference of observational studies". In: *arXiv preprint arXiv:2501.11181*.

Nie, Xinkun and Stefan Wager (2021). "Quasi-oracle estimation of heterogeneous treatment effects". In: *Biometrika* 108.2, pp. 299–319.

Robinson, Peter M (1988). "Root-N-consistent semiparametric